



Book I. Prop. V. Theor.

In any isosceles triangle  if the equal sides be produced, the external angles at the base are equal, and the internal angles at the base are also equal.

Produce  and  (post. II),
take  =  (pr. I.3);
draw  and .

Then in  and 

we have  =  (const.),

 common to both,

and  =  (hyp.)

$\therefore$  = ,  =  and
 =  (pr. I.4).

Again in  and 

we have  = ,  = 

and  = ,

$\therefore$  =  and  = 
(pr. I.4)

but  = , $\therefore$  = 
(ax. III).

Q. E. D.